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***B.Tech. Degree I Semester Regular/Supplementary and
II Semester Supplementary Examination March 2022***

CE/EE/ME/SE 19-200-0105A & CS/EC/IT 19-200-0205B

BASIC ELECTRONICS ENGINEERING

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A

(Answer *ALL* questions)

(8 × 3 = 24)

- I. (a) Explain classification of material based on energy bands?
- (b) Compare Drift and Diffusion current in semiconductors.
- (c) Why bridge rectifier circuits are preferred over center-tapped rectifier?
- (d) Explain junction capacitance associated with a diode.
- (e) Determine the DC current gain β_{dc} , α_{dc} and the emitter current I_E for a transistor where $I_B = 40 \mu A$ and $I_C = 4.5 mA$.
- (f) Briefly explain any one method to test the transistor.
- (g) Compare diffusion process and epitaxial growth.
- (h) Describe punch through effect in transistors.

PART B

(4 × 12 = 48)

- II. (a) Sketch and explain the typical voltage-current characteristics for a forward-biased silicon pn-junction. What is the change in VI characteristics if the working temperature is increased by 10°C? (4)
- (b) Define donor and acceptor doping with necessary diagram. (4)
- (c) A silicon pn-junction has a reverse saturation current of $I_o = 40 nA$ at a temperature of 300 K. Calculate the junction current when the applied voltage is (i) 0.7 V forward bias (ii) 15 V reverse bias. (4)

OR

- III. (a) A sample of silicon is doped with 3.3×10^{14} acceptor atoms/cm³ and 4×10^{14} donor atoms/cm³. Determine the resultant density of free electrons and holes in the sample. (5)
- (b) Explain the processes of electron movement and hole transfer in semiconductors. (3)
- (c) Explain the formation of depletion region in pn junction. (4)
- IV. (a) Explain the working of a Centre-tapped full wave rectifier with capacitive filter. (8)
- (b) Differentiate Zener breakdown and Avalanche breakdown. (4)

OR

- V. (a) The turns ratio of the transformer used in a bridge rectifier is $n_1 : n_2 = 12 : 1$, The primary is connected to 220 V, 50 Hz mains. Assuming that the diode voltage drops to be zero, find the dc voltage across the load and rectification efficiency. What is the PIV of each diode? (5)
- (b) Draw power supply block diagram with output waveforms. (4)
- (c) Explain the working of a Light Emitting Diode with neat diagram. (3)

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- VI. (a) With diagram explain the working of an NPN transistor. (8)
(b) Explain the working of Function Generator with suitable diagram. (4)
- OR**
- VII. (a) Explain the characteristics of BJT CB configuration with diagram. (8)
(b) How transistor works as an amplifier? (4)
- VIII. (a) Explain IC packaging with diagrams. (4)
(b) Explain the Mesa transistor fabrication technique. What is its drawback and suggest a solution in detail. (8)
- OR**
- IX. (a) Explain the process of preparing semiconductor material for device manufacturing. (5)
(b) Explain the equation for determining power gain and power level changes when two power levels are measured and when two voltage levels are measured and explain about half power points. (4)
(c) Differentiate thick film and thin film IC manufacturing. (3)
